

Three Notions of Linearization for a Discrete Ising System: Operator, Statistical, and Projection-Based Approximations

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Abstract—For stochastic, high-dimensional nonlinear systems, the notion of “linearization” is ambiguous and depends on how state, dynamics, and observables are represented. In this project, we study a discrete-time Ising model with Glauber dynamics and ask how three distinct linear representations differ in their fixed points, stability properties, and retained structure. We construct each representation from the same underlying dynamics and analyze how aggregation and coarse-graining alter equilibria and bifurcation behavior. Our analysis showcases the idea that linearization is not a single algorithm (a first-order approximation at a point) but a modeling choice that trades off between exactness, complexity reduction, and interpretability, with concrete consequences for captured behavior.

I. MOTIVATION

Nonlinear systems theory traditionally emphasizes local linearization via the Jacobian. However, for stochastic and high-dimensional systems, especially those arising in statistical physics and multi-agent decision processes, multiple nonequivalent notions of “linearization” coexist. These include exact linear representations over probability simplices, heuristic approximations of variables, and data-driven reduced-order linear models obtained by projection.

II. INTRODUCTION

This report studies multiple Ising-inspired models developed to compare several simplification strategies for stochastic, high-dimensional dynamics. The central question is not whether one can produce a reduced model, but how different modeling choices preserve or distort the behavior of a common reference system. In particular, we examine how operator-based, probability-based, and coarse-grained representations differ in their ability to reproduce transient dynamics, long-time behavior, and parameter-dependent trends.

The Ising model provides a useful setting for this comparison. It is simple enough to simulate under controlled parameter grids, yet rich enough to exhibit collective behavior, sensitivity to temperature, and substantial differences between microscopic and macroscopic descriptions. Those features make it a natural testbed for studying approximation, aggregation, and coarse-graining. At the same time, it supports a pedagogical objective: each simplification makes a different assumption about what information is essential, and the resulting mismatch between models helps clarify the consequences of those assumptions.

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All model simplifications are evaluated and compared under shared initial-spin atomic-lattice seeds, common parameter conventions, and equal trajectory horizons. The reference model [1] defines the target behavior, and the remaining models are judged by how well they reproduce its manifold of trajectories and summary statistics. This framing keeps the emphasis on model validity domains, approximation error, and the interpretation of residuals, rather than on the formal derivation of any single reduced model.

III. BACKGROUND

A. The Ising setting

We consider a discrete spin system on a graph, typically a regular lattice, with spin variables taking values in $\{-1, +1\}$. Neighbor interactions are encoded by coupling parameters J_{ij} , an external field h biases the spins globally, and temperature T controls the degree of stochasticity in the updates. Positive coupling favors alignment, negative coupling favors anti-alignment, and larger temperatures weaken the deterministic influence of the local field.

This system is useful because it exhibits qualitatively distinct regimes. In ordered regimes, aligned domains dominate and the system develops nontrivial magnetization. In disordered regimes, local correlations weaken and the state behaves increasingly like noise. Near criticality, however, the system develops structure across multiple scales, making it difficult to describe accurately using only local or only global summaries. This multiscale behavior is precisely what makes the model suitable for a study of simplification and coarse-graining.

B. Why multiple simplifications are needed

For high-dimensional stochastic systems, there is no single canonical notion of simplification. One may preserve the full probability law while changing the state representation, track only marginal probabilities, or project the system onto lower-dimensional observables such as magnetization. Each choice produces a different object: a full operator model, a structured probability update, or a coarse reduced dynamics. These are not interchangeable. They retain different information and therefore support different kinds of analysis.

That distinction is central to this report. A compact model may capture long-time magnetization while missing local transient structure. A higher-fidelity model may represent detailed state evolution but become intractable beyond small lattices. The purpose of this analysis is therefore not merely to reduce dimension, but to expose the tradeoffs between fidelity, interpretability, and computational cost.

IV. SYSTEM DESCRIPTION

We consider an N -spin Ising system with configuration $s \in \{-1, +1\}^N$

evolving in discrete time according to Glauber-type stochastic [2] [2]. For each atom i , the update probability depends on the current configuration s_i , the coupling strengths $J_{ij} \in [-1, +1]$ with neighbor j , an external field $h \in \mathbb{R}$, and a temperature $T \geq 0$. In the proposal formulation,

$$P(s_i^{k+1} = +1 | s^k) = \sigma \left(\sum_j J_{ij} s_j^k + \mathcal{N}(h, T^2) \right),$$

where $\sigma(\cdot)$ is the logistic function and $\mathcal{N}(\mu, \sigma^2)$ is a normal random variable. This defines a nonlinear stochastic map on the configuration space. The interaction structure is determined by J , while h and T control global bias and update randomness, respectively.

Although the microscopic dynamics are nonlinear, several useful surrogate descriptions can be built by changing the state representation. The remainder of the report compares five such models under a common experimental protocol.

V. MODEL SIMPLIFICATIONS

The study considers five models. Model 1, the true Ising model, serves as the canonical reference, and Models 2–5 represent different simplifications or restructurings of the same underlying dynamics. The goal is not to rank them abstractly, but to determine which features of the reference dynamics each one preserves. The code we used for simulating these models is found at [3].

A. Model 1: Reference dynamics

Model 1, the previously detailed true Ising model, is the reference model used to define the target trajectory manifold. It provides the benchmark against which the other models are evaluated. In this workflow, all translation maps, residual diagnostics, and trajectory overlays are interpreted relative to Model 1.

B. Model 2: Mean-field magnetization chain

Model 2 compresses the system to a single-state description based on global magnetization (mean spin). The dynamics on the mean magnetization m , using mean scalar coupling J , are:

$$m^{k+1} \approx \tanh(Jm^k + \mathcal{N}(h, T^2)).$$

Its main advantage is simplicity: it is fast, interpretable, and directly connected to mean-field intuition. Its main limitation is that it cannot represent local correlation structure or heterogeneous transients, particularly at the famous critical point of the Ising model. It is therefore best viewed as a global summary model rather than a detailed dynamical surrogate.

C. Model 3: Local-neighborhood probability model with mixing

Model 3 creates a coarse-grain description by incorporating neighborhood-level information through local probability updates and a mixing mechanism. Essentially, local groups of atoms induce an approximate perturbation to the mean field h , and atoms in the neighborhood react to this instead of checking each individual coupling. This gives it access to more spatial structure than Model 2 while remaining more tractable than a full configuration model. The tradeoff is that the mixing step can alter transient rates, so good long-time agreement does not necessarily imply accurate short-time dynamics.

D. Model 4: Full exponential-state Gibbs-style model

Model 4 is essentially a Monte Carlo Markov Chain and retains the richest mechanistic structure among the approximations considered here. This model lifts the dynamics to the probability of an entire lattice transitioning to different lattice configurations, so for N atoms, the model considers 2^N states. The local dynamics are given by:

$$P(s_i = +1 | \text{neighbors}) = \sigma \left(\frac{2(J \cdot \sum_{j \in \text{neighbors}} s_j + h)}{T} \right).$$

These are iterated over the entire grid and accumulated to produce a probability distribution over lattices for the next time step. It is closest in spirit to an operator-level or full-state probabilistic description, and for small lattices it can reproduce structure that simpler models necessarily average away. Its limitation is combinatorial growth in the state space, which makes it unsuitable for larger systems.

E. Model 5: Restricted-interval affine probability operator

Model 5 uses an affine operator acting on per-site probabilities under explicit interval and normalization constraints. This representation is attractive because its components are interpretable and its parameters are bounded in a controlled way. At the same time, those structural constraints can reduce expressiveness, especially in regimes where the reference dynamics are strongly nonlinear or exhibit competing transient and steady-state effects.

In this model, we represent the state of our MDP as a probability distribution over each spin being ± 1 . Each of the N spins has two associated nonnegative values, $p_i(+1) + p_i(-1) = 1$, stacked in a vector $p \in \mathbb{R}_{[0,1]}^{2N}$. This naturally partitions the following transition matrices into 2×2 blocks, each block corresponding to one atom.

We restrict or normalize the coupling, external field, and temperature to the ranges $J_{ij} \in [-1, +1]$, $h \in [-1, +1]$, $T \in [0, 1]$, respectively. The transition matrix $A(J, h, T)$ is:

$$\frac{1}{2}T\mathbf{1} + (1 - T) \left(|h| \mathbf{H}(h) + (1 - |h|) \mathbf{J}_*(J) \right).$$

Where $\mathbf{1}$ is the block-diagonal matrix with 2×2 blocks of ones on the diagonal blocks, and zeros on the off-diagonal blocks, which effectively homogenizes the probabilities of each spin to $p_i(+1) = p_i(-1) = 0.5$, $\forall i$. $\mathbf{H}(h)$ is also

a block-diagonal matrix with the same structure, but the diagonals are:

$$\begin{bmatrix} \frac{1+h}{2} & \frac{1+h}{2} \\ \frac{1-h}{2} & \frac{1-h}{2} \end{bmatrix}$$

This adds a bias to each next-timestep probability that is independent of the previous probabilities (as the row vectors homogenize the previous distribution). Finally, $\mathbf{J}_*(J)$ is a block matrix of a similar shape, but the (i, j) th 2×2 block of $\mathbf{J}_*$ is:

$$\frac{1}{\sum_k |J_{ik}|_0} \begin{bmatrix} \frac{1+J_{ij}}{2} & \frac{1-J_{ij}}{2} \\ \frac{1-J_{ij}}{2} & \frac{1+J_{ij}}{2} \end{bmatrix},$$

with zero blocks where $J_{ij} = 0$. This operator is the only of the three where the rows are heterogeneous, and they have the effect of homogenizing the distribution when $J_{ij} = 0$, biasing toward alignment (pos. larger $\rightarrow$ pos. larger, etc.) when $J_{ij} > 0$, and biasing toward anti-alignment when $J_{ij} < 0$. The normalization constant ensures that the entire $\mathbf{J}_*$ matrix is row-stochastic by dividing out the number of nonzero blocks in that block-row, i.e. $|x|_0 \triangleq \mathbb{I}(x \neq 0)$, the nonzero indicator or zero-norm. Row stochasticity is the required condition for each individual spin to always have cumulative probability one; column stochasticity is not required as it is each individual spin that requires a conservation of probability density, not the entire p vector.

The following heuristics motivate the dynamics of this simplified model:

- As $T \rightarrow 1$, the dynamics are drowned by the noise or temperature term and the expected spin is everywhere $\frac{1}{2}$.
- Else, as $h \rightarrow \pm 1$, the external field overwhelms the spins, and the expected spins tend toward ± 1 .
- Else, if the mean coupling is positive, spins will tend to be more like the average of their neighbors. Likewise, if the mean coupling is negative, spins will tend to be opposite the average of their neighbors.

VI. EXPERIMENTAL PROTOCOL

To compare the models fairly, all experiments use shared lattice sizes where tractable, matched seeded initial conditions or seed-derived initial distributions, common parameter grids over (J, h, T) (though the arguments to some models were scaled to better align the steady-states of each), and consistent trajectory horizons. The resulting outputs are summarized on a common manifold indexed by model, coupling, field, temperature, and time, with the height of the manifold being the mean-field magnetization. This structure allows direct comparison of trajectory means, variability, and fit residuals without conflating model differences with changes in experimental setup.

From the simulated trajectories, we form grouped summaries with index

$$(\text{model}, J, h, T, t),$$

and statistics including the sample mean magnetization, variance, and sample count. These summaries support both

qualitative figure-based comparison and quantitative residual analysis.

VII. CROSS-MODEL PARAMETER TRANSLATION

To compare models on equal footing, we fit translation maps from the parameter space of one model to the manifold behavior of another. The first stage uses nearest-neighbor matching between source and target summaries. The second stage fits affine approximations to those correspondences, producing explicit cross-model parameter maps. This translation step makes it possible to ask not only whether two models can match, but where and how they fail to match.

In selected cases, manual parameter adjustments were introduced in place of the automated fit. These adjustments are treated as explicit interventions rather than hidden tuning. Their purpose is to improve trajectory alignment and act as a change of units to make comparing the models more intuitive. Specifically, we scaled the temperature T of Models 2 and 3 by a factor of 1.5, the external field strength h of Model 5 by 8, and increased the number of dynamics updates per time step to 6 in Model 3 and 8 in Model 4. These adjustments were made to make the evolutions in Fig. 1 align better.

VIII. FIGURE GUIDE AND INTERPRETATION

A. Summary-grid plot

Figure 1 provides the most direct visual summary of the model hierarchy. Each row fixes a parameter setting (J, h, T) (with manual adjustments of units, as mentioned above), and each column shows the corresponding behavior under one model. The central curve is the mean magnetization trajectory, while the shaded region represents a 95% confidence interval. This figure shows cross-sections of the trajectory manifold. Note that the behavior is fairly consistent across models, with slight differences in variance and steady-state bias. This figure was generated using a 2×3 atom lattice with consistent initial configurations across model and parameter settings.

IX. DISCUSSION

The comparisons indicate that the main difficulty in cross-model alignment is not simply poor parameter fitting. In many regimes, a single affine map must account for three distinct effects: stochastic variability, transient convergence, and steady-state magnetization.

The inherent variance of some of the models considered is a particularly noteworthy finding. Specifically, in Fig. 1, we see that Model 4 exhibits more variability in certain parameter regimes than the other models. Not pictured in Fig. 1 due to size limitations, Models 2 and 3 also have poor variance when $h = 0$, though all models do find the correct magnetization of 0 given these uniform initial configurations.

The transient behavior also varies from model to model. Models 3 and 4 had slower convergence rates, so the adjustment in the number of updates brought their curves into a more similar shape to the others. We also see that some

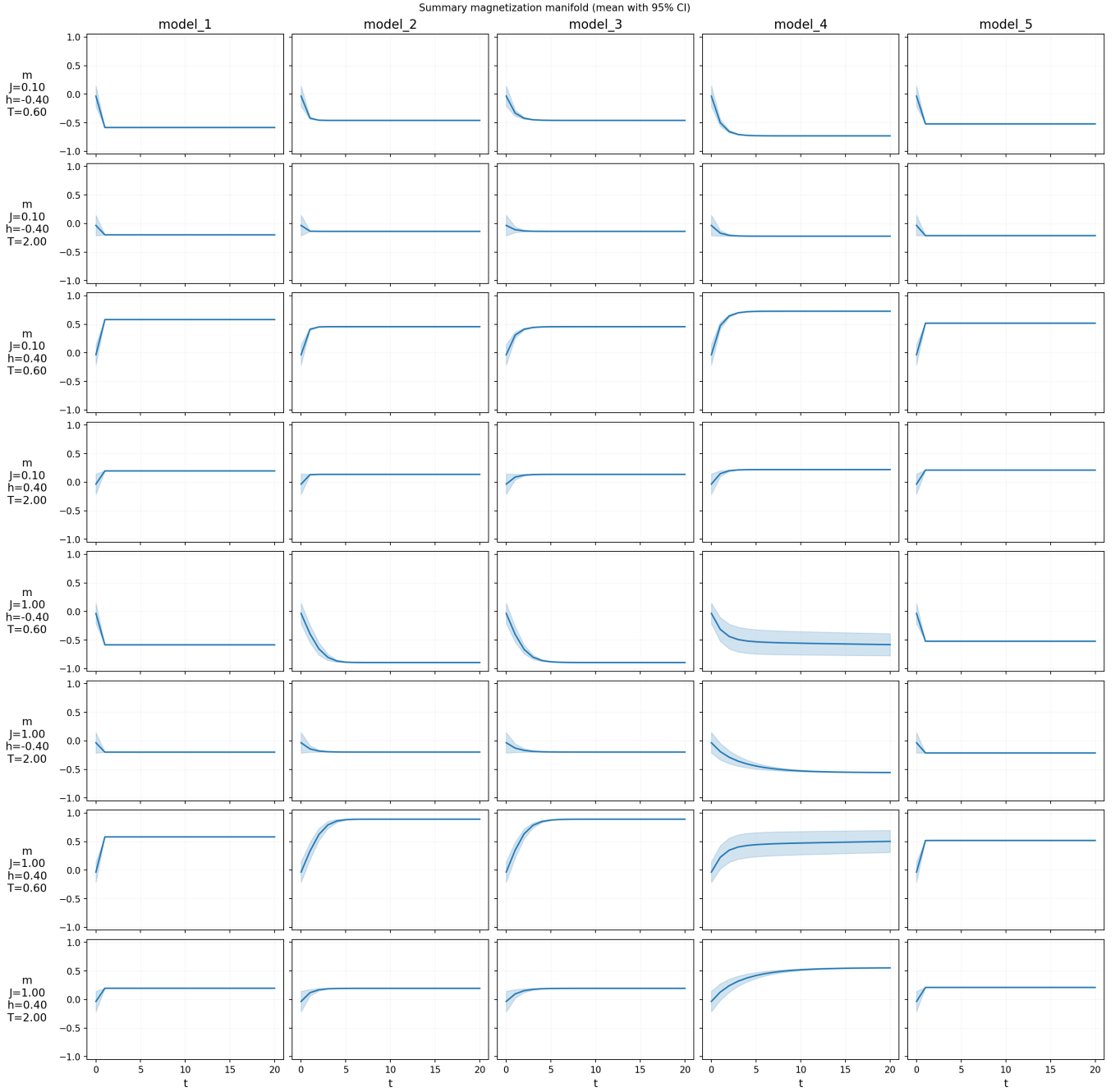


Fig. 1. Summary manifold panels. Columns correspond to models and rows correspond to parameter settings (J, h, T) . Each panel shows the mean magnetization trajectory $m(t)$ together with a 95% confidence interval. Not all parameter regions are shown here.

transients (Models 1 and 5) are sharper, while the others (particularly Model 4) are more gradual.

The steady state was also tricky to make consistent between models. These correspond to equilibria in the magnetization dynamics. One shortcoming of this experiment was evaluation time and the length of the report; with more compute, we should include initial conditions with biased proportions of spins, and initial clusters of similar spins of varying sizes, repeated across many more parameter settings.

While these effects stem from different structural features of the models, a global set of parameters (J, h, T, t) seems

to be sufficient to describe the model behavior, once the units for particular models have been adjusted. This explains why some tuning can improve visual agreement between the trajectories in Fig. 1. This is also good evidence that these concepts are central to the heart of the behaviors the Ising model captures, or in other words, that these parameters are sufficient statistics to describe Ising-type systems. As sufficient statistics, we can use them in our simplified models to exact useful data for predicting the mean-field magnetization of the lattice without needing to compute the effect of each individual atom. The sufficient statistics form an information

Markov blanket between the dynamics of the full spin model and the dynamics of the magnetization.

X. CONCLUSION

The five-model comparison studied here provides an analysis of simplification strategies for Ising-type stochastic dynamics. The comparison is useful both mathematically and pedagogically. Mathematically, it clarifies how operator-level, probability-level, and coarse-grained models differ in fidelity, tractability, and error structure. Pedagogically, it shows that disagreement between models is informative: residuals, failed mappings, and projection-order effects identify exactly which assumptions matter and where a simplification stops being trustworthy.

The main contribution of the report is therefore methodological. A good simplification is not the one that appears most elegant in isolation, but the one whose assumptions, retained structure, and failure modes are all made explicit.

A. Future Work

A natural next step is to separate the calibration of transient and steady-state behavior. One possibility is to use a genetic algorithm to find optimal linear transforms on the parameters (including frequency of dynamics updates), with Model 1 as the baseline comparison. The hand-tuning I performed assumed a diagonal transform, but this provided adequate results for our demonstration.

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APPENDIX A: EMERGENCE, EXPLANATION, AND UNIVERSALITY

This appendix is a philosophical reflection on emergence, abstraction, and universality, inspired by [4]. The central claim is that emergence is not merely a mysterious property

of complicated systems, but is closely tied to abstraction, model compression, and the development of languages that can express increasingly general kinds of structure. In that sense, the philosophy of emergence is also a philosophy of explanation.

A. Competing Hypotheses and Better Explanations

You cannot disprove a hypothesis in isolation; there must be some set of hypotheses competing. Relativity did not simply replace Newtonian mechanics on its own. Rather, Newtonian mechanics was confronted not only with new data, but with additional assumptions, such as the implicit idea that light should travel in straight lines, and it was the combination of those assumptions that made the old theory inadequate.

Science can answer what, when, where, and how, but it cannot finally answer why. Two people can agree on the mechanisms that set in motion the Big Bang, supersymmetry breaking, accretion, nuclear fusion, organic chemistry, information storage and retrieval, evolution, universal language, enlightenment, industrialization, and even the AI singularity, and still disagree completely on why any of it exists. Science looks at both of them and says only: "I cannot tell why."

Still, some explanations are better than others. Natural language explanations are easily varied, like stories, while the precise nature of constructive languages such as mathematics is less so. A mathematical model can still be varied to explain data, but there is a difference between changing the story of Persephone and changing the weights in a neural network. The deeper issue is whether a model merely summarizes data or gives a mechanistic account with greater reach.

Kepler summarized planetary motion geometrically, but could not explain why that geometry existed. Newton explained that geometry through an inverse-square force, but could not explain where that force came from. Einstein explained that bodies follow geodesics in curved spacetime, but still did not explain in any ultimate sense why energy curves spacetime. Each explanation kicked the can of explanation further down the road, but each was still better than the last. A deeper explanation should imply the shallower one, should make new predictions that can be experimentally validated, and should replace more variable or "magical" axioms with less variable ones. In this sense, mechanistic models have better reach: when a model continues to work in novel situations without being constantly adjusted, it is more likely to match the actual mechanism of the system.

B. Emergence as Abstraction and Compression

Emergence is the quality of a system composed of components that interact according to a set of rules, and whose composition exhibits behavior that the individual components cannot exhibit [5]. In other words, each component does not work as a good approximate model of the entire system. These systems often exist at a critical point between too few rules and too many rules, such as Conway's Game of Life [6] and chess on an infinite board, which are both Turing-complete. The author's opinion is that this property of crit-

icality (which often does manifest as Turing-completeness) is essential to sufficiently interesting systems and plays a complementary role in the philosophy of modeling such systems by intelligent agents.

Discussion on emergence from structured composition is fundamentally a discussion about scale. The phenomenon of emergence is often accompanied by a Markov blanket of scale; in fact, the entire philosophy of reductionism stems from the assumption that large and small scales are separated by a Markov blanket of only a few key features. For example, the properties of a gas made up of moles of atoms are described best not by the quantum field theory governing individual atoms, but by the volume the atoms fill, the number of atoms, the mean kinetic energy of the atoms (temperature), and the frequency of collisions, including with the boundaries of the volume (pressure). These quantities are related by an ideal gas law, which is much more tractable and descriptive of the gas as a whole than describing the states of each individual atom. This process of aggregating small-scale features into higher-scale laws is formalized in *renormalization* [7], [8], [9]. This approach goes by many names, including projection, course-graining, renormalization, emergence, abstraction, aggregation, mean-field theory, sufficient statistics, balanced truncation model reduction [10], Koopman observables [11], [12], [13], feature augmentation, or simply “stepping back,” “zooming out,” or “seeing the big picture.”

Emergence may be understood as the segmentation, or possibly nonlinear change of basis, on data that scales and is adequate to describe higher-order features with maximal specificity. Said differently, emergence is a form of model compression. For example, one could attempt to model the phase of water using the number of molecules, the maximum kinetic energy among molecules, and the average number of molecular collisions per second. But the more effective three-parameter compression uses pressure, temperature, and volume. In that sense, those variables are emergent.

The number of emergent parameters creates a cliff-plateau structure in model accuracy: increased complexity can give increased accuracy, but only at a cost. For example, we could create a model of water phases that accounts for the expected number of nucleation sites, which might improve accuracy or may simply hamper calculations without much of an accuracy payout. The cliffs are critical points where emergence allows efficient compression. The emergent features are those that carry the most information into the abstraction, and so they provide the most efficient description of the higher-level model. More specifically, they are aggregation functions: we cluster particles by mean kinetic energy, or customers by the items they purchase. The levels of these aggregation functions build levels of expressivity, and the critical cliffs at each scale indicate the importance of the aggregated features.

This produces a criticism of reductionism. In one sense, there is no downward causation in a hierarchy of abstraction; features aggregate upward, and lower-level systems cause higher-level effects. Yet designed systems complicate this picture. There are cases in which explaining the behavior of

the parts seems to require explaining the design of the whole, not merely enumerating the components. It is not enough to say which atoms are present in a statue or a computer; one must also explain why they are arranged as they are.

In this sense, there are at least two useful notions of emergence. The first is simple emergence: combinations of states or variables that carry information into higher-level abstractions, such as temperature or the categories into which people cluster groceries. The second is complex emergence: the appearance of systems whose structure begins to approach universality.

C. Knowledge Under Limited Computation

Knowledge is time-limited information. All of the information may already be contained in the data, constraints, and objective, but the amount of information that can actually be accessed depends on computation. The order in which one discovers abstractions matters when computation is limited. If useful abstractions are discovered early, a process can progress faster and farther within its computational budget.

For example, in Conway’s Game of Life, the sooner one discovers gliders, the sooner one can reason about the system at a higher level and avoid drowning in cell-by-cell detail. Similarly, if one were building an operating system, it would be disastrous to reason only at the bit level and postpone functional abstraction until the very end; the computation would never finish. Abstractions create knowledge because they trade some local accuracy for tractable computation. The more knowledge one has, the faster one can replicate or extend the data, though leaky abstractions can introduce error.

Grokking has this same character. At first, one struggles through a family of related problems by brute force. Then one discovers a unifying approach that summarizes and generalizes the earlier effort. Once the abstraction is found, the formerly difficult problems become almost trivial. In this sense, learning is often the discovery of the right abstraction.

D. Universality, Syntax, and Criticality

Complex emergence is closely related to universality. The jump is from a finite, specialized architecture to a mutable, effectively unbounded memory capable of encoding arbitrarily many architectures. Crystals are self-replicating finite automata; DNA-based life is something closer to self-replicating Turing machines. Life is not merely self-replicating information. It is mutable, unbounded memory.

Human language also has this universal character. The human is the head, and the string of sounds or symbols is the tape. More deeply, universality is not just the existence of symbols, but the acceptance of syntax as the carrier of meaning rather than the symbols themselves. In analog systems, a new symbol must often be invented for each new meaning. In universal systems, a relatively small number of building blocks, together with rules of syntax and composition, generate an enormous expressive space.

This is why constraints can deconstrain. Accepting a standard can enable universality. The only thing better than

perfect is standardized. Shipping containers are a simple example: universal packaging increases efficiency by orders of magnitude. Likewise, mathematics and music are taught poorly when students merely memorize special cases. A more universal education teaches the discrete components and the syntax under which those components can be composed. Then one learns not a specific equation or a specific song, but a language for producing many equations or many songs.

Universality also seems to live near a critical point. Too much structure in a syntax and the system becomes rigid; too little and it loses expressive power. Roman numerals fail as a universal numerical language not because their symbols are inadequate in principle, but because extraneous constraints limit concatenation. Likewise, Conway's Game of Life appears near a knife-edge: too few or too many rules and it loses Turing completeness. Interesting systems often seem to exist between under-constraint and over-constraint.

There is also a distinction between analog and digital communication here. Human beings communicate with universal language, but also with analog forms such as music, faces, smells, and images (animal calls are also analog). A picture may be worth a thousand words precisely because it accesses memory and emotion without passing through a fully universal syntax. Music has many ingredients of a universal language, such as the composition of discrete notes, but lacks a standardized syntax that fixes the meaning of every phrase. It was developed, in some sense, to communicate motion and emotion rather than propositional content.

Human civilization appears to be another layer of emergence above the individual. No single human builds an airplane, and arguably no single company or even nation does so in complete isolation. Civilizational intelligence is a real thing: recorded language, theory, science, standardization, and division of labor create a superstructure in which knowledge accumulates. The part of the brain repurposed for reading unlocked a universality beyond spoken language, a universality at the level of civilization.

E. Intelligent Agents and Universal Modeling

Intelligent agents are those that have developed a universal computational framework for processing, modeling, and controlling the exponential state spaces of emergent phenomena. We use language to describe the complexity of the world, mathematics to describe structures generated by a few rules, and strategies and algorithms to navigate enormous combinatorial spaces such as chess or the Rubik's cube. Universal intelligence is what allows sparse and enormous state spaces to become tractable.

This suggests one way to think about learning systems. Reinforcement learning and Q-learning may be understood as attempts to form manageable abstractions of exponential state spaces. Likewise, current language models may still fall short of general intelligence because they have not fully made the jump to universality. They often treat tokens, token strings, or even the entire context window as specialized latent objects. If such systems were given a more explicit

way to manipulate dynamic memory and operate over compositional symbolic structure, they might move closer to universality.

The same may be true of the modeling problem itself. Perhaps our language of model architectures is not yet universal. We often treat each architecture as a specialized object with its own attached meaning. A more universal language of models would push meaning upward into the composition layer, so that individual modules become less important than the syntax by which they are assembled.

Any explanation of the output of a universal computer must therefore include an explanation of the syntax used to formulate the problem, perform the computation, and interpret the solution.

F. Models, Misclassification, and Social Meaning

Model failures are not interpreted in a vacuum. A misclassification is always also interpreted through social and historical context. For that reason, two statistically identical model failures may be treated very differently by society depending on the narratives layered on top of them. This does not make the modeling failure less real, but it does mean that the meaning of the failure is not exhausted by statistical description alone.

That observation matters for philosophy of modeling because it reminds us that explanation operates at more than one level. A model may fail as a classifier, but the public meaning of that failure may reflect broader human failures, historical patterns, or institutional biases. The model is still making an error, but the interpretation of that error depends on a larger system of human abstraction and social meaning.

G. Humanity, Beauty, and Civilizational Intelligence

One would have to be a fool not to recognize that humans are unique and special. We see beauty where we never should have. We see beauty not only in flowers that evolved for the attraction of pollinating insects, but in distant galaxies and nebulae that no creature on Earth was ever meant to see directly, often visible only after false-color reconstruction. We take cold dust on the horizon of an otherwise empty universe and call it majestic. In this sense, we create beauty as much as we discover it.

This is both a blessing and a curse. The same intelligence that gives meaning and joy to existence also makes us aware of mortality. The universe does not mourn death as we do; death is the natural state of the universe. And yet our ability to recognize impermanence is precisely what gives so much of human life its beauty. We connect moments across space and time through memory, nostalgia, grief, and love. The universe itself does not make those connections. We do.

There is some real change when matter becomes an intelligent agent. We are no longer tied only to the optimization process that shaped our evolution. Something else emerges: a universal intelligence that can model quasars, invent flight instead of waiting for it to evolve, breed species for arbitrary traits, preserve history, formulate explanations, and act collectively through institutions and recorded knowledge. In that

sense, humanity is not merely another phenomenon within nature, but one of nature's clearest examples of emergence through universality.